UPDATE FOR MISSING DISCUSSION POINTS

In your Gold Trails Manual, a page was altered calling Granite Sedimentary with a short incomplete explanation. Below is the original text. Please print this and insert it into your book.

This was included due to a recent find of sedimentary gold bearing granite with no explanation on how this can happen by the Miner.

Granite: From Igneous Rock to Sedimentary Rock

How independent placer gold can enter the deposit

The granite starts as igneous rock

Granite forms when magma cools slowly beneath the surface. Its interlocking crystals, commonly quartz, feldspar and mica, make it an intrusive igneous rock. That classification describes how the original rock formed. It does not tell us what may eventually happen to the material when weathering and erosion break the granite apart.

Weathering is not the same as becoming sedimentary rock

Exposed granite weathers. Feldspar may alter to clay, fractures widen, and the rock can break down into grains and fragments known as decomposed granite, or grus. If the material remains where it formed, it is still weathered granite. Loose decomposed granite is not automatically a sedimentary rock just because it looks like sand and gravel.

The change happens when that material is transported and deposited

Water, gravity and sometimes wind move the loose material away from its parent rock. As it is transported, the grains can become mixed, sorted, rounded and deposited in washes, channels, fans or basins. At this point we have sediment derived from granite. With burial, compaction, and natural cementation, that sediment can harden into a new sedimentary rock. Feldspar-rich sand commonly forms arkose, a type of sandstone. Coarse rounded granite fragments may be cemented into conglomerate; angular fragments may form sedimentary breccia.

Here is the important distinction: an individual granite pebble within a conglomerate remains igneous granite. The rock made of deposited and cemented pebbles is sedimentary. Granite supplied the ingredients, but the new rock is classified by the process that formed it.

Now introduce placer gold from an independent source

Imagine a granite mountain that contains no significant gold. An ancient stream erodes it, carrying the resulting quartz, feldspar and granite fragments downstream. That same stream may also collect placer gold from an entirely different mineralized area farther upstream or from an older deposit it cuts through. When the gold and granite-derived sediment settle together, they occupy the same deposit even though they did not come from the same rock.

If that gravel and sand are later buried and cemented, they can become an ancient gold-bearing conglomerate or sandstone. The gold is then preserved in what geologists may call a paleoplacer: an old placer deposit. Later erosion can release the gold again and produce a younger placer. None of this requires the local granite to have supplied the gold.

What does this mean when I am prospecting?

First, identify what you are standing on: intact granite, decomposed granite still in place, loose transported sediment, or cemented sedimentary rock. Then look for evidence of the water that deposited the material: old channels, layering, changes in gravel size, and places where heavy minerals could concentrate. Finding gold in granite-derived sediment tells you where gold was deposited, but by itself it does not identify where that gold originally formed.

Although rare this is where the event is typically found
Granite - coarse fragments - conglomerate or breccia

Granite - sand-sized grains - sandstone

If the sediment contains lots of relatively fresh feldspar from rapidly eroded granite, it can form arkose, essentially a feldspar-rich sandstone strongly associated with granitic source rock.

Granite can also weather all the way down into clay minerals, which can eventually contribute to shale or mudstone.

And here's the part that matters in the field: you can absolutely find pieces of granite locked inside a sedimentary rock. A conglomerate or breccia, for example, might contain rounded granite cobbles that were eroded out of an older granite body, carried by water, deposited, and then cemented together.This also holds true for the deposition and concentration of gold.

So geologically, the sequence could be:

Granite - weathering/erosion - sediment - deposition - burial/cementation - sedimentary rock.

Granite can supply the sediment without supplying the gold.